

Lab Worksheet: Parent–Child CASCADE with SELECT (MySQL)

Objective	SQL Query	Expected Output
Create and select database	DROP DATABASE IF EXISTS demo_cascade; CREATE DATABASE demo_cascade; USE demo_cascade;	Database created and selected.
Create Parent table (AUTO_INCREMENT)	CREATE TABLE Parent (ParentID INT AUTO_INCREMENT PRIMARY KEY, ParentName VARCHAR(50) NOT NULL);	Parent table created.
Create Child table with CASCADE (AUTO_INCREMENT)	CREATE TABLE Child (ChildID INT AUTO_INCREMENT PRIMARY KEY, ParentID INT NOT NULL, ChildName VARCHAR(50), CONSTRAINT fk_child_parent FOREIGN KEY (ParentID) REFERENCES Parent(ParentID) ON UPDATE CASCADE ON DELETE CASCADE) ENGINE=InnoDB;	Child table created with FK CASCADE.
Insert Parent rows	INSERT INTO Parent (ParentName) VALUES ('Parent A'), ('Parent B');	2 rows inserted into Parent.
Insert Child rows	INSERT INTO Child (ParentID, ChildName) VALUES (1, 'Child A1'), (1, 'Child A2'), (2, 'Child B1');	3 rows inserted into Child.
Check initial data	SELECT * FROM Parent; SELECT * FROM Child;	Parent → IDs 1 & 2. Child → linked rows (A1,A2,B1).
CASCADE UPDATE: change ParentID=1 → 10	UPDATE Parent SET ParentID = 10 WHERE ParentID = 1;	Child rows with ParentID=1 auto-updated to 10.
Check after cascade update	SELECT * FROM Parent; SELECT * FROM Child;	Parent has IDs 10 & 2. Child A1,A2 updated to ParentID=10.

CASCADE DELETE: remove ParentID=2	DELETE FROM Parent WHERE ParentID = 2;	Parent row 2 deleted. Child with ParentID=2 auto-deleted.
Check after cascade delete	SELECT * FROM Parent; SELECT * FROM Child;	Parent has ID=10 only. Child A1,A2 remain, Child B1 deleted.
Insert new Child for ParentID=10	INSERT INTO Child (ParentID, ChildName) VALUES (10, 'Child A3');	Child A3 inserted under ParentID=10.
Delete a Child manually	DELETE FROM Child WHERE ChildID = 1;	Child with ID=1 deleted. Parent unaffected.
Insert new Parent and reassign Child	INSERT INTO Parent (ParentName) VALUES ('Parent C'); UPDATE Child SET ParentID = 3 WHERE ChildID = 2;	Child 2 reassigned to new ParentID=3.
Final check	SELECT * FROM Parent; SELECT * FROM Child;	Parent → IDs 10 & 3. Child → linked correctly.
SELECT all rows	SELECT * FROM Parent; SELECT * FROM Child;	Shows all columns and rows.
SELECT specific columns	SELECT ParentID, ParentName FROM Parent; SELECT ChildID, ChildName FROM Child;	Shows only chosen columns.
SELECT with WHERE clause	SELECT * FROM Parent WHERE ParentID = 10; SELECT * FROM Child WHERE ParentID = 10;	Filters rows matching condition.
SELECT with ORDER BY	SELECT * FROM Parent ORDER BY ParentName ASC; SELECT * FROM Child ORDER BY ChildName DESC;	Sorted output (ascending/descending).
Aggregate SELECT (GROUP BY)	SELECT ParentID, COUNT(*) AS TotalChildren FROM Child GROUP BY ParentID;	Shows number of children per parent.